The intensity (I) which is proportional to the square of the amplitude can be written in the form

$$I = 4I_0 \cos^2\phi/2 \qquad \textbf{(14.13)}$$

where I_0 is the intensity produced by each one of the source individually. Clearly if $\phi = \pm\pi, \pm 3\pi, \ldots$, the resultant intensity will be zero and we will have minima. On the other hand, when $\phi = 0, \pm 2\pi, \pm 4\pi, \ldots$, the intensity will be maximum ($= 4I_0$). However, if the phase difference between the sources S_1 and S_2 (i.e., ϕ) is changing with time, the observed intensity will be given by

$$I = 4I_0 \left\langle \cos^2 \frac{\phi}{2} \right\rangle \qquad \textbf{(14.14)}$$

where $\langle \ldots \rangle$ denotes the time average of the quantity inside the angular brackets; the time average of a time dependent function is defined by the following relation:

$$\langle f(t) \rangle = \frac{1}{\tau} \int_{-\tau/2}^{+\tau/2} f(t)\, dt \qquad \textbf{(14.15)}$$

where τ represents the time over which the averaging is carried out. For example, if the interference pattern is viewed by a normal eye, this averaging will be over about 1/10th of a second; for a camera with exposure time 0.001 sec, $\tau =$ 0.001 sec. etc. Clearly, if ϕ varies in a random manner in times which are small compared to τ, then $\cos^2 \phi/2$ will randomly vary between 0 and 1 and $< \cos^2 \phi/2 >$ would be 1/2 [see also Sec. 14.6]. For such a case,

$$I = 2I_0 \qquad \textbf{(14.16)}$$

which implies that if the sources are incoherent then the resultant intensity is the sum of the two intensities and there is no variation of intensity! Thus, if one (or both) of the two vibrating sources are turned on and off in a random manner (such that the phase difference between the vibrations of the two sources varies rapidly) then the interference phenomenon will not be observed. We will discuss this point again in Sec. 14.6 and also in Chapter 17.

14.4 INTERFERENCE OF LIGHT WAVES

LO 3

Till now we have considered interference of waves produced on the surface of water. We will now discuss the interference pattern produced by light waves; however, for light waves it is difficult to observe a stationary interference pattern. For example, if we use two conventional light sources (like two sodium lamps) illuminating two pinholes (see Fig. 14.5), we will not observe any interference pattern on the screen. This can be understood from the following reasoning: In a conventional light source, light comes from a large number of independent atoms; each atom emitting light for about 10^{-10} sec, i.e., light emitted by an atom is essentially a pulse lasting for only 10^{-10} seconds* Even if the atoms were emitting under similar conditions, waves from different atoms would differ in their initial phases.

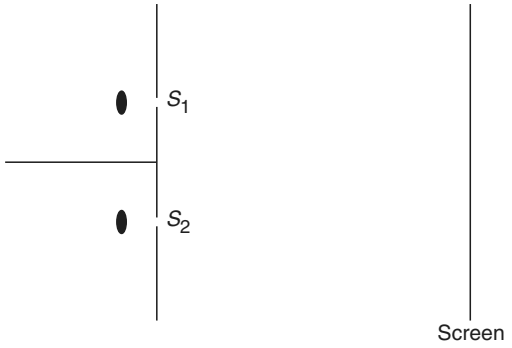


Fig. 14.5 If two sodium lamps illuminate two pinholes S_1 and S_2, no interference pattern will be observed on the screen.

Consequently, light coming out from the holes S_1 and S_2 will have a fixed phase relationship for a period of about 10^{-10} seconds, hence the interference pattern will keep on changing every billionth of a second. The eye can notice intensity changes which last at least for a tenth of a second and hence we will observe a uniform intensity over the screen. However, if we have a camera whose time of shutter opening can be made less than 10^{-10} seconds then the film will record an interference pattern**. We summarise the above results by noting that light beams from two independent sources do not have any fixed phase relationship***, as such they do not produce any stationary interference pattern.

Thomas Young in 1801 devised an ingenious but simple method to lock the phase relationship between the two sources. The trick lies in the division of a single wavefront into two; these two split wavefronts act as if they emanated from two sources having a fixed phase relationship and,

* Since the optical frequencies are of the order of 10^{15} s^{-1}, such a short pulse consists of about a million oscillations; thus it is almost monochromatic (see Chapter 17).

** This interference pattern will be a set of dark and bright bands only if the light waves have the same state of polarization. This can, however, be easily done by putting two polaroids in front of S_1 and S_2. We should mention here that by using two independent laser beams it has been possible to record the interference pattern (see Chapter 17).

*** Such sources are termed as incoherent sources.

therefore, when these two waves were allowed to interfere, a stationary interference pattern was obtained. In the actual experiment, a light source illuminates the pinhole S (see Fig. 14.6). Light diverging from this pinhole fell on a barrier which contained two pinholes S_1 and S_2 which were very close to one another and were located equidistant from S. Spherical waves emanating from S_1 and S_2 (see Fig. 14.7) were coherent and on the screen beautiful interference fringes were obtained. In order to show that this was indeed an interference effect, Young showed that the fringes on the screen disappear when S_1 (or S_2) is covered up. Young explained the interference pattern by considering the principle of superposition, and by measuring the distance between the fringes he calculated the wavelength. Figure 14.7 shows the section of the wavefront on the plane containing S, S_1 and S_2 (which is the $x - z$ plane).

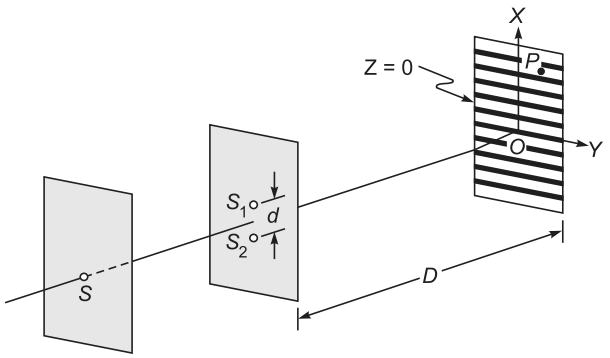


Fig. 14.6 Young's arrangement to produce interference pattern.

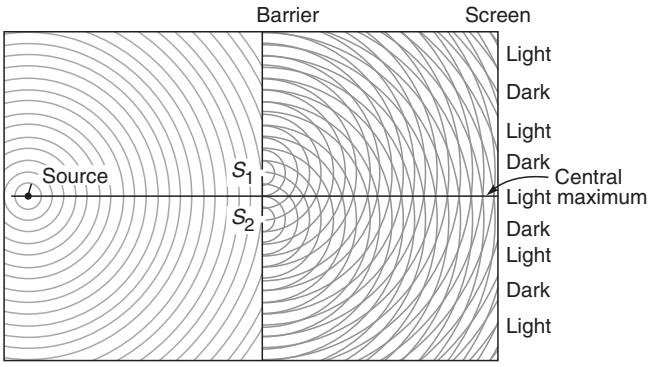


Fig. 14.7 Sections of the spherical wavefronts emanating from S, S_1 and S_2 (Adapted from Ref. 14.7; used with permission).

14.5 THE INTERFERENCE PATTERN

LO 4

In this section, we will first obtain an expression for the fringe width and then we will show that the fringes are strictly hyperbolic.

Let S_1 and S_2 represent the two pinholes of the Young's double hole arrangement. We would determine the positions of maxima and minima on the line LL' which is parallel to the x axis and lies in the plane containing S, S_1 and S_2 (see Fig. 14.8). We will show that the interference pattern (around the point O) consists approximately of a series of dark and bright lines perpendicular to the plane of Fig. 14.8; O being the foot of the perpendicular from the point S on the screen.

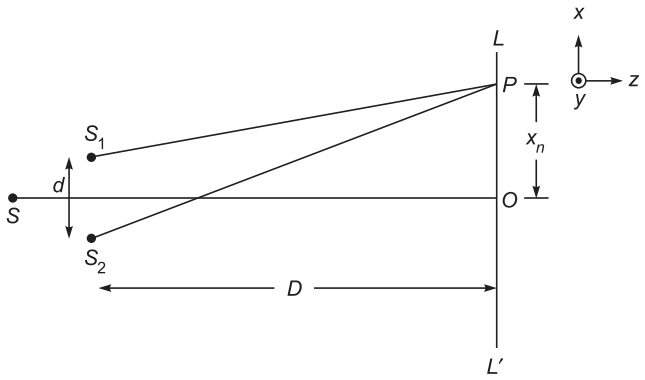


Fig. 14.8 Arrangement for producing Young's interference pattern.

For an arbitrary point P (on the line LL') to correspond to a maximum we must have

$$S_2P - S_1P = n\lambda;\ n = 0, 1, 2, 3,\ldots \qquad \textbf{(14.17)}$$

Now,

$$(S_2P)^2 - (S_1P)^2 = \left[D^2 + \left(x_n + \frac{d}{2}\right)^2\right] - \left[D^2 + \left(x_n - \frac{d}{2}\right)^2\right] \qquad \textbf{(14.18)}$$

$$= 2x_nd$$

where $S_1S_2 = d$ and $OP = x_n$. Thus,

$$S_2P - S_1P = \frac{2x_nd}{S_2P + S_1P} \approx \frac{x_nd}{D} \qquad \textbf{(14.19)}$$

where in the last step we have replaced $S_2P + S_1P$ by $2D$ which will be valid when $D >>> d, x_n$. For example, for $d =$ 0.02 cm, $D = 50$ cm, and $OP = 0.5$ cm (which corresponds to typical values in a light interference experiment) we will have

$$S_2P + S_1P = \sqrt{(50)^2 + (0.51)^2} + \sqrt{(50)^2 + (0.49)^2}$$

$$= 100.005 \text{ cm}$$

Thus, if we replace $S_2P + S_1P$ by $2D$, the error involved is about 0.005%. Using Eqs. (14.17) and (14.19) we obtain

$$x_n = \frac{n\lambda D}{d} \tag{14.20}$$

Thus, the bright (and dark) fringes are equally spaced and the distance between two consecutive bright (or dark) fringes is given by

$$\beta = x_{n+1} - x_n = \frac{\lambda D}{d} \tag{14.21}$$

which is the expression for the fringe width.

We will next determine the shape of the interference pattern on the screen LL' and show that the fringes are a set of hyperbolae. We assume the origin to be at the point O and the z-axis to be perpendicular to the plane of the screen LL' as shown in Fig. 14.8. The screen LL' corresponds to the plane $z = 0$; thus the coordinates of an arbitrary point P on the screen will be $(x, y, 0)$. The coordinates of the point S_1 and S_2 would be $\left(+\frac{d}{2}, 0, -D\right)$ and $\left(-\frac{d}{2}, 0, -D\right)$, respectively. Thus,

$$S_2P - S_1P = \left[\left(x + \frac{d}{2}\right)^2 + y^2 + D^2\right]^{1/2} - \left[\left(x - \frac{d}{2}\right)^2 + y^2 + D^2\right]^{1/2}$$

$$= \Delta \text{ (say)}$$

or

$$\left(x + \frac{d}{2}\right)^2 + y^2 + D^2 = \left\{\Delta + \left[\left(x - \frac{d}{2}\right)^2 + y^2 + D^2\right]^{1/2}\right\}^2$$

Simple manipulations will give us

$$(d^2 - \Delta^2)x^2 - \Delta^2 y^2 = \Delta^2\left[D^2 + \frac{1}{4}(d^2 - \Delta^2)\right]$$

For $\Delta = 0$ (zero path difference) we must have $x = 0$ which implies that the central (bright) fringe is along the y-axis; this is rigorously true. In general, the above equation can always be written in the form

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \tag{14.22}$$

where

$$a^2 = \frac{\Delta^2}{(d^2 - \Delta^2)}\left[D^2 + \frac{1}{4}(d^2 - \Delta^2)\right]$$

and

$$b^2 = D^2 + \frac{1}{4}(d^2 - \Delta^2) \tag{14.23}$$

Equation (14.22) represents a hyperbola. On rearranging, we get

$$x = \pm\left(\frac{\Delta^2}{d^2 - \Delta^2}\right)^{1/2}\left[y^2 + D^2 + \frac{1}{4}(d^2 - \Delta^2)\right]^{1/2} \tag{14.24}$$

Obviously for $y^2 << D^2$, we may neglect y^2 inside the square brackets and the loci are straight lines parallel to the y-axis. Thus, we obtain straight line fringes on the screen. We must remember that we had assumed point sources and we obtained straight line fringes. It is easy to see that if we had slits instead of point sources, each pair of points would have produced the same straight line fringes which would have overlapped with each other–thus we would again obtain straight line fringes. The fringes so produced are said to be non-localized; they can be photographed by just placing a film on the screen; they can also be seen through an eyepiece.

14.6 THE INTENSITY DISTRIBUTION

LO 5

Let $\mathbf{E}_1$ and $\mathbf{E}_2$ be the electric fields produced at the point P by S_1 and S_2 respectively (see Fig. 14.8). The electric fields $\mathbf{E}_1$ and $\mathbf{E}_2$ will, in general, have different directions and different magnitudes. However, if the distances S_1P and S_2P are very large in comparison to the distance S_1S_2, the two fields will almost be in the same direction. Thus, we may write

$$\left.\begin{aligned}\mathbf{E_1} &= \hat{\mathbf{i}}E_{01}\cos\left(\frac{2\pi}{\lambda}S_1P - \omega t\right)\\ \text{and}\quad \mathbf{E_2} &= \hat{\mathbf{i}}E_{02}\cos\left(\frac{2\pi}{\lambda}S_2P - \omega t\right)\end{aligned}\right\} \tag{14.25}$$

where $\hat{\mathbf{i}}$ represents the unit vector along the direction of either of the electric fields. The resultant field will be given by

$$\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2 = \hat{\mathbf{i}}\left[E_{01}\cos\left(\frac{2\pi}{\lambda}S_1P - \omega t\right) + E_{02}\cos\left(\frac{2\pi}{\lambda}S_2P - \omega t\right)\right] \tag{14.26}$$

The intensity (I) will be proportional to the square of the electric field and will be given by

$$I = KE^2 \tag{14.27}$$

or

$$I = K\left[E_{01}^2\cos^2\left(\frac{2\pi}{\lambda}S_1P - \omega t\right) +$$

$$E_{02}^2 \cos^2\left(\frac{2\pi}{\lambda} S_2P - \omega t\right) +$$

$$E_{01}\,E_{02}\left\{\cos\left[\frac{2\pi}{\lambda}(S_2P - S_1P)\right] +$$

$$\cos\left[2\omega t - \frac{2\pi}{\lambda}(S_2P + S_1P)\right]\right\}\Bigg] \qquad \textbf{(14.28)}$$

where K is a proportionality constant.* For an optical beam the frequency is very large ($\omega \approx 10^{15}$ sec^{-1}) and all the terms depending on ωt will vary with extreme rapidity (10^{15} times in a second); consequently, any detector would record an average value of various quantities. Now,

$$\left\langle \cos^2(\omega t - \theta)\right\rangle = \frac{1}{2\tau}\int_{-\tau}^{+\tau} \frac{1+\cos[2(\omega t-\theta)]}{2}\,dt$$

$$= \frac{1}{2} + \frac{1}{16\pi}\frac{T}{\tau}\left\{\left[\sin 2(\omega t-\theta)\right]_{-\tau}^{+\tau}\right\}$$

where $T = \frac{2\pi}{\omega}$ ($\approx 2\pi \times 10^{-15}$ sec for an optical beam). For any practical detector** $\frac{T}{\tau} <<< 1$ and since the quantity between the curly brackets will always be between –2 and +2, we may write

$$\langle \cos^2(\omega t - \theta)\rangle \approx \frac{1}{2} \qquad \textbf{(14.29)}$$

The factor $\cos(2\omega t - \phi)$ will oscillate between +1 and –1 and its average will be zero as can indeed be shown mathematically. Thus the intensity, that a detector will record, will be given by

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\delta \qquad \textbf{(14.30)}$$

where

$$\delta = \frac{2\pi}{\lambda}(S_2P - S_1P) \qquad \textbf{(14.31)}$$

represents the phase difference between the displacements reaching the point P from S_1 and S_2. Further

$$I_1 = \frac{1}{2}KE_{01}^2$$

represents the intensity produced by the source S_1 if no light from S_2 is allowed to fall on the screen; similarly $I_2 = \frac{1}{2}KE_{02}^2$ represents the intensity produced by the source S_2 if no light from S_1 is allowed to fall on the screen. From Eq. (14.30) we may deduce the following:

(a) The maximum and minimum values of $\cos\delta$ are +1 and –1, respectively; as such the maximum and minimum values of I are given by

$$\left.\begin{aligned} I_{\max} &= (\sqrt{I_1} + \sqrt{I_2})^2 \\ I_{\min} &= (\sqrt{I_1} - \sqrt{I_2})^2 \end{aligned}\right\} \qquad \textbf{(14.32)}$$

and

The maximum intensity occurs when

$$\delta = 2n\pi, \quad n = 0, 1, 2, \ldots$$

or

$$S_2P \sim S_1P = n\lambda,$$

and the minimum intensity occurs when

$$\delta = (2n+1)\pi\,;\, n = 0, 1, 2, \ldots$$

or

$$S_2P \sim S_1P = \left(n + \frac{1}{2}\right)\lambda$$

Notice that when $I_1 = I_2$, the intensity minimum is zero. In general, $I_1 \neq I_2$ and the minimum intensity is not zero.

(b) If the holes S_1 and S_2 are illuminated by different light sources (see Fig. 14.5), then the phase difference δ will remain constant for about 10^{-10} sec (see discussion in Sec. 14.3) and thus δ would also vary with time*** in a random way. If we now carry out the averaging over time scales which are of the order of 10^{-8} sec, then

$$\langle \cos\delta\rangle = 0$$

and we obtain

$$I = I_1 + I_2$$

Thus, for two incoherent sources, the resultant intensity is the sum of the intensities produced by each one of the sources independently and no interference pattern is observed.

(c) In the arrangement shown in Fig. 14.6, if the distances S_1P and S_2P are large in comparison to d, then

$$I_1 \approx I_2 = I_0 \text{ (say)}$$

and

$$I = 2I_0 + 2I_0\cos\delta = 4I_0\cos^2\frac{\delta}{2} \qquad \textbf{(14.33)}$$

---

* Equation (14.27) will be derived in Sec. 23.5. In free space, the constant K will be shown to be equal to $\varepsilon_0 c^2$ where ε_0 (=8.854 × 10^{-12} Coul^2/N–m^2) represents the permittivity of free space and c, the speed of light in free space.

** For a normal eye, $\tau \approx 0.1$ s; thus $T/\tau \approx 6 \times 10^{-14}$; even for a detector having 1 nsec as the resolution time, $T/\tau \approx 6 \times 10^{-5}$.

*** Notice that this variation occurs in times of the order of 10^{-10} sec which is about a million times longer than the times for variation of the intensity due to the terms depending on ωt. Thus we are justified in first carrying out the averaging which leads to Eq. (14.30).

The intensity distribution (which is often termed as the $\cos^2$ pattern) is shown in Fig. 14.9. The actual fringe pattern (as it will appear on the screen) is shown in Fig. 14.10. Figures 14.10 (a) and (b) correspond to d = 0.005 mm ($\beta \approx 5$ mm) and d = 0.025 mm ($\beta \approx 1$ mm), respectively. Both figures correspond to D = 5 cm and $\lambda = 5 \times 10^{-5}$ cm. The values of the parameters are such that one can see the hyperbolic nature of the fringe pattern in Fig. 14.10(a).

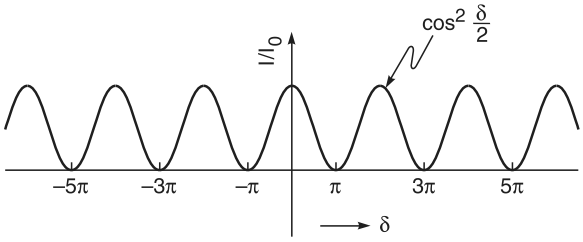


Fig. 14.9 The variation of intensity with δ.

Example 14.5 Instead of considering two point sources, we consider the superposition of two plane waves as shown in Fig. 14.11(a). The wave vectors for the two waves are given by

$$\mathbf{k}_1 = -\hat{\mathbf{x}}\ k \sin\theta_1 + \hat{\mathbf{z}}\ k \cos\theta_1$$

and

$$\mathbf{k}_2 = +\hat{\mathbf{x}}\ k \sin\theta_2 + \hat{\mathbf{z}}\ k \cos\theta_2$$

where $k = 2\pi/\lambda$ and θ_1 and θ_2 are defined in Fig. 14.11(a). Thus the electric fields of the two waves are described by the following two equations:

$$E_1 = E_{01} \cos [\mathbf{k_1} \cdot \mathbf{r} - \omega t]$$
$$= E_{01} \cos [-k x \sin\theta_1 + k z \cos\theta_1 - \omega t]$$
$$E_2 = E_{02} \cos [\mathbf{k_2} \cdot \mathbf{r} - \omega t]$$
$$= E_{02} \cos [k x \sin\theta_2 + k z \cos\theta_2 - \omega t]$$

where we have assumed both electric fields along the same direction (say along the y-axis); if we further assume $E_{01} = E_{02} = E_0$ and $\theta_1 = \theta_2 = \theta$ then the resultant field will be given by

$$E = 2E_0 \cos (k x \sin\theta) \cos (k z \cos\theta - \omega t)$$

Thus, the intensity distribution on the photograph plate LL' will be given by

$$I = 4 I_0 \cos^2(k x \sin\theta)$$

and the fringe pattern will be strictly straight lines (parallel to the y-axis) with fringe width given by

$$\beta = \frac{\lambda}{2 \sin\theta}$$

Figure 14.11(b) shows the computer generated interference pattern on the screen LL' for $\theta = \pi/6$ and λ = 5000 Å. Thus $\beta = \lambda$ = 0.0005 mm.

Example 14.6 In this example, we once again consider the interference pattern produced by 2 point sources S_1 and S_2 on a plane PP' which is perpendicular to the line joining S_1 and S_2 [see Fig. 14.12(a)]. Obviously, on the plane PP', the locus of the point P for which

$$S_1P - S_2P = \text{constant}$$

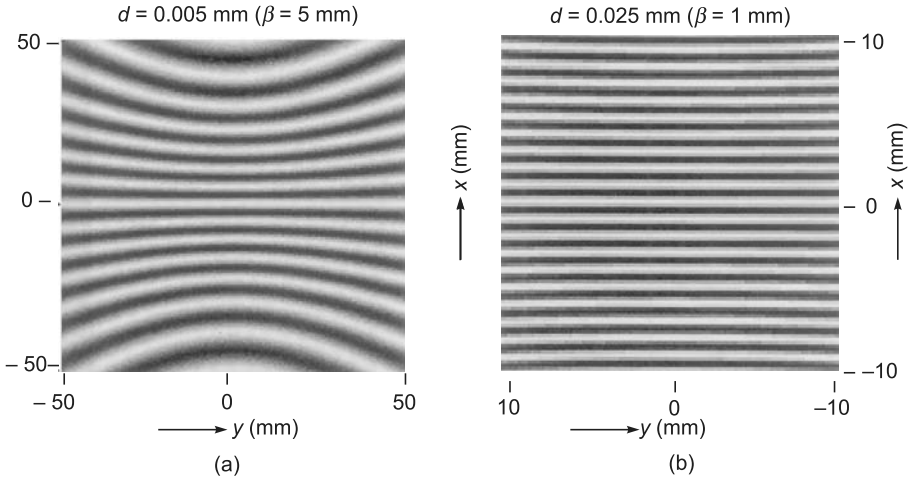


Fig. 14.10 Computer generated fringe pattern produced by two point sources S_1 and S_2 on the screen LL' (see Fig. 14.8); (a) and (b) correspond to d = 0.005 mm and 0.025 mm, respectively (both figures correspond to D = 5 cm and $\lambda = 5 \times 10^{-5}$ cm).

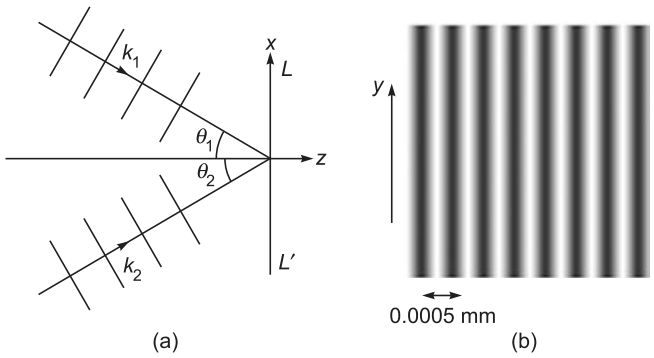


Fig. 14.11 (a) The superposition of two plane waves on LL'. (b) Computer generated interference pattern on the screen LL' for $\theta_1 = \theta_2 = \pi/6$ and $\lambda = 5000$ Å. The fringes are parallel to the y-axis.

will be a circle. Figures 14.12(b) and (c) show the fringe patterns for $D = 20$ cm and $D = 10$ cm; for both figures $S_1S_2 = d = 0.05$ mm and $\lambda = 5000$ Å. Obviously, if O represents the centre of the fringe pattern then

$$S_1O - S_2O = d = 100\ \lambda$$

Thus (for this value of d) the central spot will be bright for all values of D and will correspond to $n = 100$. The first and second bright circles will correspond to a path difference of 99 λ and 98 λ, respectively. Similarly, the first and second dark rings in the interference pattern will correspond to a path difference of 99.5 λ and 98.5 λ, respectively. The radii of the fringes can be calculated by using the formula given in Problem 14.10.

Example 14.7 We finally consider the interference pattern produced on PP' by the superposition of a plane wave incident normally and a spherical wave emanating from the point O (see Fig. 14.13). The plane wave will be given by

$$E_1 = E_0 \cos(kz - \omega t + \phi)$$

and the spherical wave will be given by

$$E_2 = \frac{A_0}{r} \cos(kr - \omega t)$$

where r is the distance measured from the point O which is assumed to be the origin. Now, on the plane PP' ($z = D$)

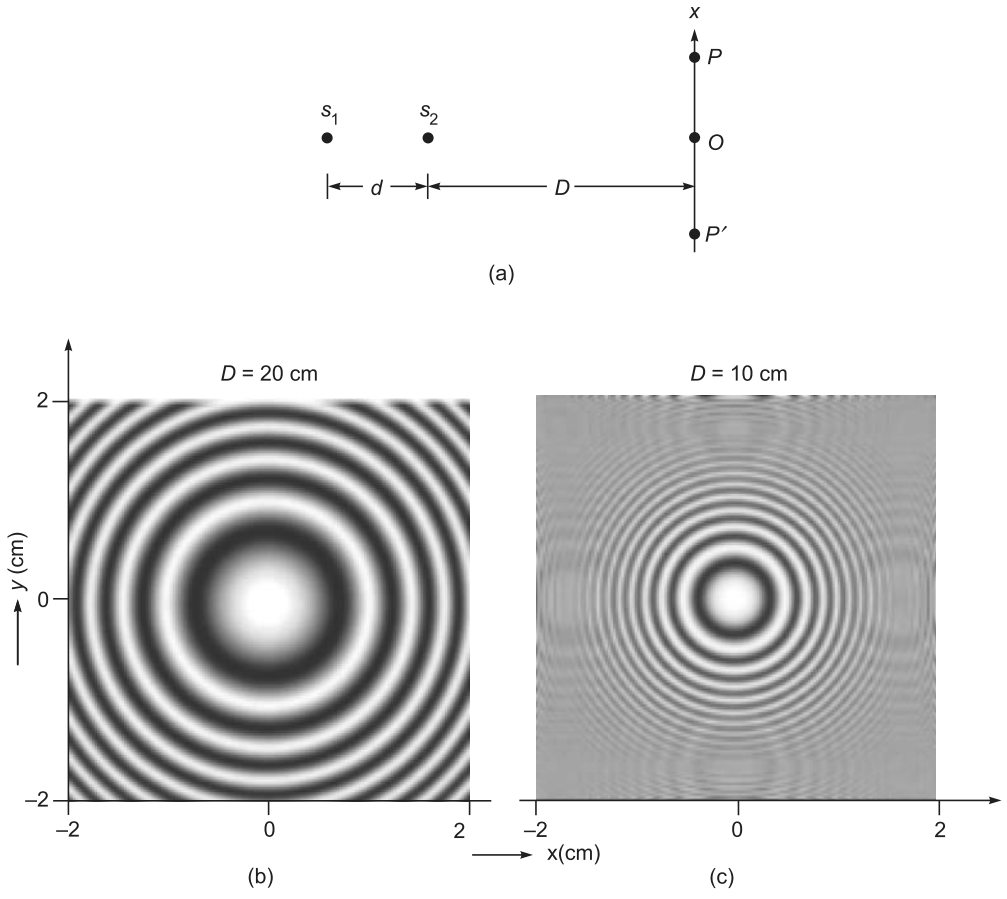


Fig. 14.12 (a) S_1 and S_2 represent two coherent sources, (b) and (c) show the interference fringes observed on the screen PP' when $D = 20$ cm and $D = 10$ cm, respectively.

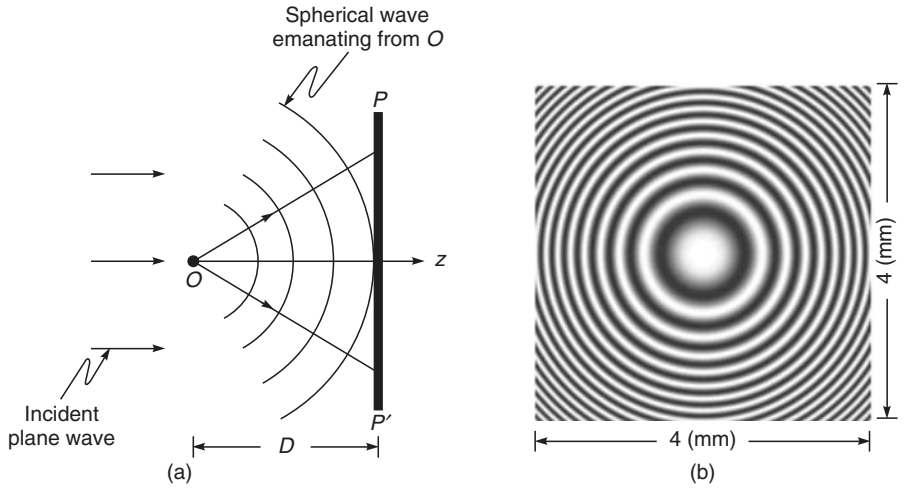


Fig. 14.13 (a) Superposition of a plane wave and a spherical wave emanating from the point O; (b) shows the interference fringes observed on the screen PP'.

$$r = (x^2 + y^2 + D^2)^{\frac{1}{2}} \approx D\left[1 + \frac{x^2 + y^2}{2D^2}\right]$$

$$\approx D + \frac{x^2 + y^2}{2D}$$

where we have assumed $x, y << D$. On the plane $z = D$, the resultant field will be given by

$$E = E_1 + E_2$$
$$\approx E_0 \cos(kD - \omega t + \phi) + \frac{A_0}{D}\cos\left[kD + \frac{k}{2D}(x^2 + y^2) - \omega t\right]$$

Thus,

$$\left\langle E^2 \right\rangle = \frac{1}{2}E_0^2 + \frac{1}{2}\left(\frac{A_0}{D}\right)^2 + E_0\frac{A_0}{D}\cos\left[\frac{k}{2D}(x^2 + y^2) - \phi\right]$$

If we assume that

$$\frac{A_0}{D} \approx E_0$$

i.e., the amplitude of the spherical wave (on the plane PP') is the same as the amplitude of the plane wave, then

$$\left\langle E^2 \right\rangle \approx 2E_0^2 \cos^2\left(\frac{k}{4D}(x^2 + y^2) - \frac{1}{2}\phi\right)$$

and we would obtain circular interference fringes as shown in Fig. 14.13(b). If r_m and r_{m+p} denote the radii of mth and $(m + p)^{\text{th}}$ bright ring, then

$$r_{m+p}^2 - r_m^2 = 2p\lambda D$$

14.6.1 Moire Fringes

We may mention here that Moire fringes can be very effectively used to study the formation of fringe patterns. In Fig. 14.14, we have shown the overlapping of two simple patterns from which one can understand the formation of bright and dark fringes when two plane waves propagate in slightly different directions. In a classroom, it can be easily demonstrated by having a periodic pattern on a transparency and overlapping it with its own photocopy at different angles. Similarly, if one overlaps a circular pattern (on a transparency) with its own copy, one obtains the hyperbolic fringes as shown in Fig. 14.15. (To get a clearer fringe pattern, you may have to view the patterns from a greater distance.) In Sec. 17.5, we have shown how the beat phenomenon can be understood by observing the Moire fringes obtained by the overlapping of two patterns of slightly different periods (see Fig. 17.13).

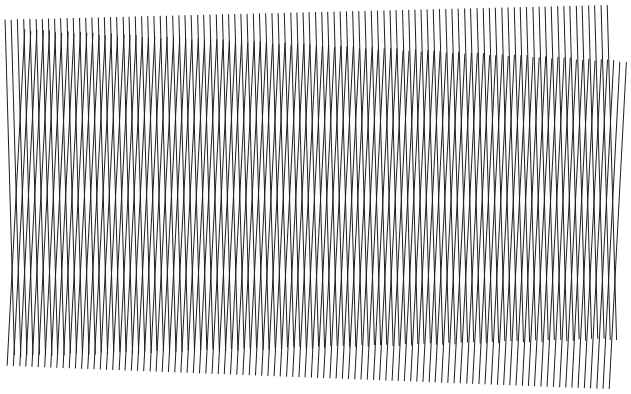


Fig. 14.14 The Moire Pattern produced by two overlapping straight line patterns.

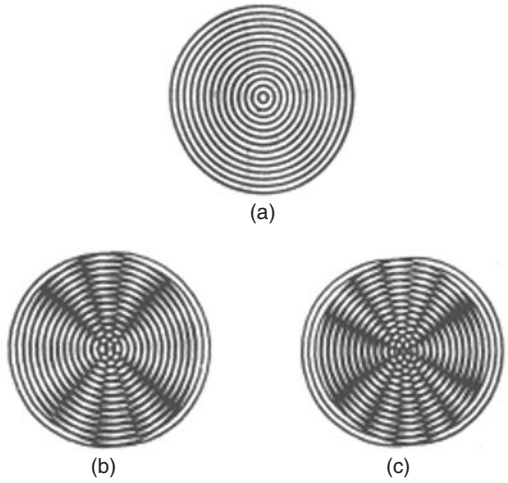


Fig. 14.15 The Moire pattern produced by two overlapping circular patterns. You will see clear hyperbolic fringes if you put the pattern at a greater distance from the eye. The circular pattern was provided by Dr. R. E. Bailey.

Example 14.8 Consider a parallel beam of light (from a distant source S' like a star incident at an angle θ) on two slits S_1 and S_2 as shown in Fig. 14.16. Obviously, the path difference between the waves emanating from the slits S_1 and S_2 will be given by

$$X S_2 = d \sin \theta$$

Therefore, the intensity distribution on the screen due to S' will be given by

$$I = I_0 \cos^2 \frac{\delta}{2}$$

where

$$\delta = \frac{2\pi}{\lambda}[XS_2 + S_2P - S_1P] = \frac{2\pi}{\lambda}[S_2P - S_1P) + d \sin \theta]$$

$$= \frac{2\pi}{\lambda}\left[\frac{xd}{D} + d \sin \theta\right]$$

Thus, the intensity distribution (due to light coming from the distant source S') will be given by

$$I' = I_0 \cos^2\left(\frac{\pi}{\lambda}\left[\frac{xd}{D} + d \sin \phi\right]\right)$$

Similarly, if there is light incident from another distant source S'' (at an angle ϕ) then the corresponding intensity distribution on the screen will be given by

$$I'' = I_0 \cos^2\left(\frac{\pi}{\lambda}\left[\frac{xd}{D} - d \sin \phi\right]\right)$$

The resultant intensity distribution will be given by

$$I = I' + I''$$

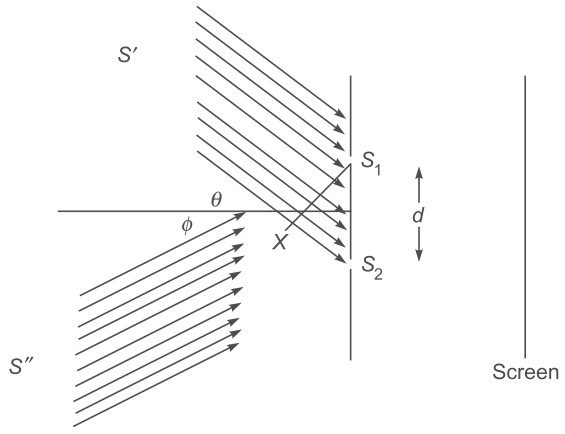


Fig. 14.16 Two distant incoherent sources S' and S'' illuminate the slits S_1 and S_2.

Example 14.9 This example presupposes the knowledge of half-wave plates (see Sec. 22.6) and therefore the reader may skip this example until he has gone through Chapter 22.

Consider a y-polarized beam light beam incident on a double hole system as shown in Fig. 14.17. Behind the hole S_1 we have put a half-wave plate H_1 whose optic axis is along the y direction and behind the hole S_2 we have put a half-wave plate H_2 whose optic axis is along the x direction. Thus, as discussed in Sec. 22.6, in H_1, a y-polarized beam will propagate with velocity c/n_e and in H_2, a y-polarized beam will propagate with velocity c/n_o. In calcite $n_e < n_o$; and, in a half-wave plate a phase change of π is introduced between the o-wave and the e-wave. Thus, the whole fringe pattern will shift by $\beta/2$ where β is the fringe width.

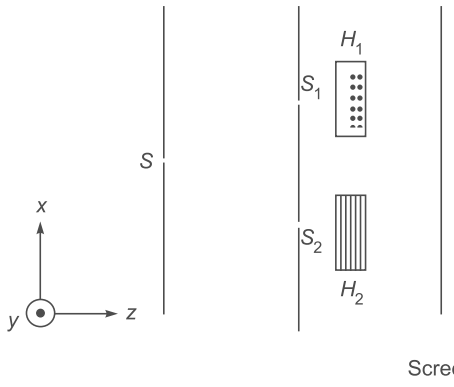


Fig. 14.17 H_1 and H_2 are half-wave plates placed in front of the slits S_1 and S_2. The optic axis of H_1 and H_2 are along y and x directions, respectively.